

Group-aware validation collapses extraction-method classification in cannabis chemical data

Arthur Gonçalves Breguez

Independent researcher

ORCID: 0009-0005-8551-731X

Correspondence: arthurbreguez@gmail.com

Preprint. Code and evidence logs: <https://github.com/ArtBreguez/cannabis-extraction-ds>

Abstract

Published models that classify cannabis products from cannabinoid and terpene profiles report near-perfect accuracy: above 95% for resin type, 100% for chemovar class. These figures are obtained with random cross-validation or Kennard-Stone partitioning over data with group structure, where replicate clones, repeat producers and repeat laboratories span the train/test boundary.

We quantify the cost of that choice on a task where the group structure is explicit and measurable. Using 37,374 cannabis certificates of analysis, almost all of them concentrates, with a producer-declared extraction-method label, a gradient-boosted classifier separates solventless from hydrocarbon extracts at 81.0% balanced accuracy under random five-fold cross-validation. Holding out the producer drops it to 54.0%, a 95% confidence interval that contains the 50.0% chance level. Restricted to the 53 producers making both classes, so a held-out producer's label cannot be read off training rows, it gives 54.6%.

The collapse replicates on an independent, designed experiment: 162 HPLC-assayed samples across six varieties and three laboratory extraction methods fall from 56.7% under random folds to 38.9% under leave-one-variety-out, against 33.3% chance. In both datasets the chemistry predicts the confounder better than the target. The profile identifies the testing laboratory 15.3 points above its baseline in market data, and the plant variety at 60.7% against a 16.7% chance level in the controlled experiment.

We conclude that extraction method is not recoverable from the analyte panels in public cannabis COAs for an unseen producer. We did not rerun the published studies and make no claim about their chemical conclusions, only that their figures come from validation leaving group structure intact, the regime in which we measure 27 points of inflation.

Keywords: cannabis, chemometrics, data leakage, group cross-validation, extraction method, certificates of analysis, negative result

1. Introduction

Cannabis chemical profiles are now routinely fed to supervised classifiers. Reported performance is consistently high. A support vector machine separates Dutch-type from Moroccan-type cannabis resin at 95.2% accuracy from two cannabinoid concentrations [1]. Partial least squares discriminant analysis as-

signs chemovar class from near-infrared spectra of whole inflorescences with "100% prediction accuracy", sensitivity and specificity both 1.00 [2]. A combined cannabinoid-and-terpene PLS-DA model reaches 0% prediction misclassification across chemovars [3].

These numbers invite a question the papers do not ask: performance on *what* unseen unit? A shuffled fold defines "unseen" as a row absent from the training partition. Deployment defines it more strictly, as a plant, a producer, or a laboratory the model has never encountered. When several rows descend from the same entity, a random split lets the model recognise the entity instead of learning the property of interest, and the reported score measures memorisation.

This failure mode is established in the general methodological literature [4, 5, 6, 7], with the earliest quantitative demonstration in genomics more than two decades ago [11], and has been described specifically for chemometrics, where Kiraly and Toth show that the Kennard-Stone selection algorithm and the inappropriate handling of repeated measurements both introduce train/test leakage [8]. The cannabis applications cited above use precisely these procedures. Reference [2] grows 88 genotypes cloned in triplicate to obtain 264 plants, then partitions with Kennard-Stone [8] without grouping the clones; reference [1] draws a random 80% for training and evaluates with plain accuracy on a 149/40 class split, where always answering the majority class already scores 78.8%.

The bridge between the two literatures has not been built. No published cannabis study that we could locate holds out the producer, the genotype or the laboratory and reports what happens to the score.

We build that bridge on a concrete task. The question is whether a concentrate's cannabinoid and terpene profile carries enough information to recover how it was made, separating solventless extracts (rosin, live rosin, hash rosin) from hydrocarbon extracts (BHO, live resin, shatter, badder, wax). The task is a good test case for three reasons. The label is declared at source rather than inferred. The group structure is explicit, since every sample names a producer and a laboratory. And a substantial subset of producers make both classes, which permits a test where producer identity is provably uninformative.

We report a negative result for the chemical question and a quantitative result for the methodological one.

2. Materials and methods

2.1 Data

Market data. The Cannlytics `cannabis_results` corpus [9] aggregates public cannabis certificates of analysis obtained from state regulatory portals and published COAs, released under CC-BY-4.0. The retrieved file contains 808,407 rows, of which 139,714 are concentrates or extracts.

Labels come from the `product_type` field, which in some markets declares the extraction family directly as `non-solvent based concentrate` or `solvent based concentrate`. This is a source-declared label, not an inference from a commercial product name. Rows whose `product_type` is only `concentrate` remain unlabelled and are never assigned to a class.

Most concentrate rows carry no such declaration and are therefore dropped at this step, which is where the bulk of the reduction happens:

concentrate or extract rows	139,714	
product_type declares the extraction family	37,310	(26.7%)
product_type does not	102,404	(73.3%)
labelled rows outside that subset	64	
total labelled rows	37,374	

The discarded majority is dominated by values that name a format rather than a method: `extracts` (34,034 rows), `concentrate`, `product inhalable` (30,586), bare `concentrate` (26,030) and `marijuana extract for inhalation` (3,901). These are not missing labels to be recovered, since the extraction family was never recorded, so imputing one would invent the dependent variable. Matching is by regular expression (`non[- ]?solvent`) rather than exact string, which also admits regional variants such as `non-solvent concentrate` (1,445 rows) and, in 64 cases, `infused pre-rolls (non-solvent)`. Those 64 rows are the reason the labelled total slightly exceeds the declared count within the concentrate subset; they are solventless by declaration but are pre-rolls rather than concentrates, and they are 0.17% of the labelled data.

A secondary labelling path applies regular expressions to `product_name` (`hash rosin`, `live rosin`, `rosin` for solventless; `live resin`, `bho`, `shatter`, `badder`, `wax` for hydrocarbon). Where both paths produce a label they agree on 4,254 of 4,284 rows, 99.3%. The 30 contradictory rows are excluded.

The resulting labelled dataset is 37,374 rows, of which 30 carry contradictory labels and are excluded from every analysis, leaving **37,344 rows** that all figures below are computed on unless stated otherwise:

hydrocarbon	24,573	65.8%
solventless	12,771	34.2%

Grouping variables are populated as follows:

producer	33,229 rows (89.0%)	96 distinct
lab	37,344 rows (100.0%)	12 distinct
strain_name	0 rows (0.0%)	0 distinct

Controlled data. Zenodo record 13823859 [10] is a designed experiment rather than market data: six named varieties, each extracted by three laboratory methods (maceration, ultrasound, supercritical CO₂) under varied time, temperature and pressure, with cannabinoids quantified by HPLC. The design is fully balanced at 54 samples per method and 27 per variety, 162 total, with four features (CBD, CBG, CBN, THC) and no missing values.

2.2 Feature construction

The source file carries 35 analyte columns. Sixteen are empty for every labelled row and are dropped rather than imputed, since imputing a never-populated column fabricates a feature. Nineteen analytes remain.

Non-detects require care. The corpus has already normalised `ND` and `<LOQ` strings away, so no such strings survive, which collapses a four-way distinction into a two-way one:

- a numeric **zero** is a real measurement at or below the detection limit;

- an **empty** cell is an absence of information, either not tested or not reported.

These are not the same and must not be merged. Each analyte therefore contributes two columns: the value, and a boolean `_tested` mask. Rows are never dropped for having gaps.

This distinction produces two different and both meaningful coverage figures, which we separate explicitly because conflating them is easy:

analyte	tested (flag)		detected (value > 0)	
	solventless	hydro	solventless	hydro
d_limonene	75.7%	81.1%	40.7%	65.6%
beta_caryophyllene	75.7%	81.1%	47.0%	68.3%
alpha_pinene	75.7%	81.1%	34.2%	56.3%
total_terpenes	88.2%	89.4%	56.8%	74.5%

Terpene panels are requested at similar rates for both classes, a gap of 1 to 6 points. The large gap, 18 to 25 points, is in *detection*: hydrocarbon extracts report a non-zero terpene value far more often. That is a property of the extracts or of their reporting, not of which test was ordered.

2.3 The leakage pre-check

A detection gap of 18 to 25 points is a candidate shortcut. If the pattern of which terpenes were reported, or reported as non-zero, tracked the class, a classifier could score well while reading the reporting convention rather than the chemistry. We measured this before modelling rather than assuming it away.

A classifier given **only the missingness pattern**, with every measured value discarded, reaches 65.8% accuracy against a 65.8% majority-class baseline: a lift of **+0.0 points**. Adding laboratory identity as an explicit feature reaches 67.8%, a lift of 2.0 points.

Laboratory and class are also not entangled. Only 373 rows, 1.0%, sit in laboratories that are more than 90% one class. The two largest laboratories, 15,035 and 12,539 rows, both sit close to the overall 65.8/34.2 split.

The cheapest shortcuts therefore buy almost nothing, and any model that performs well must do so on measured chemistry. This is what made the main experiment worth running.

2.4 Model and split schemes

A single model is used throughout: `HistGradientBoostingClassifier` (scikit-learn), `max_iter=200`, fixed seed, early stopping disabled. The point of the study is the evaluation protocol, not the estimator, so the estimator is held constant across all schemes.

Four split schemes are compared on the market data:

1. **random**: stratified five-fold, the protocol used in the cannabis literature;
2. **unseen-producer**: `GroupKFold` grouped by producer, so no producer appears in both train and test;
3. **unseen-lab**: leave-one-lab-out. Each fold holds out one entire laboratory. Only laboratories with at least 500 rows are eligible to be held out, because a fold testing on a 100-row laboratory estimates nothing; 8 of the 12 laboratories clear that floor and all 8 contain both classes, giving 8 folds. The 4 excluded laboratories hold 561 rows, 1.50% of the data, and they are still present in every training set;

4. **dual-class + unseen**: restricted to the 53 producers that make *both* classes, grouped by producer.

Scheme 4 is the strongest available evidence, though not for the reason a first reading suggests. Restricting to producers who make both classes removes the degenerate case where a producer's identity determines its label outright, but it does not make producer identity uninformative: the solventless share across these 53 producers ranges from 0.00 to 0.99 with a median of 0.35, and 30 of the 53 sit outside the 0.2 to 0.8 band. A classifier given *only* an encoded producer ID, and no chemistry at all, still reaches 84.4% balanced accuracy inside this subset under a random split. What the restriction guarantees is that the held-out producer's label cannot be inferred from having seen that same producer in training, since every producer in the subset contributes both classes. It covers 27,751 rows, 74.3% of the usable data, with a 17,312 / 10,439 class split.

On the controlled data, random five-fold is compared against leave-one-variety-out.

2.5 Metrics

The classes are 1:2, so plain accuracy is not reported. We use balanced accuracy (chance 50.0% for two classes, 33.3% for three), macro F1, per-class recall, confusion matrices, and the standard deviation across folds. Two baselines are computed explicitly: most-frequent (balanced accuracy 50.0%) and stratified random (50.7%).

Per-class recall matters here because a single aggregate figure hides which side of the problem the model actually solves.

2.6 Alarm models

Before the main task, two probes asked whether the chemistry predicts metadata it should not: the testing laboratory and the producer. A profile that identifies the laboratory is a profile that partially encodes instrumentation, calibration and reporting convention, which would compromise any score obtained while train and test share laboratories. A third target, cultivar, could not be probed and is treated separately in 4.3.

Each probe keeps only classes with enough rows to be estimable, since a class with 20 members contributes noise rather than signal: at least 100 rows for the laboratory probe and 150 for the producer probe, the latter higher because producers are far more numerous. These floors, applied after the 30 conflicting rows are dropped, are why the probes in 3.1 run on 11 of 12 laboratories (37,276 rows) and 33 of 96 producers (30,416 rows) rather than on the full dataset. The `*_tested` flags are excluded from these probes, so that "the chemistry identifies the laboratory" is not conflated with "the missingness pattern identifies the laboratory", which 2.3 measures separately.

3. Results

3.1 The alarm fires first

target	classes	rows	accuracy	baseline	lift	
lab	11	37,276	55.6%	40.3%	+15.3 pts	[ALARM]
producer	33	30,416	24.3%	14.4%	+9.9 pts	[watch]

The chemical profile does identify the testing laboratory, 15.3 points above its baseline. In hindsight this is unsurprising, since instruments, calibration and reporting conventions differ between laboratories. Its consequence is not optional: any score obtained while train and test share laboratories is partly measuring the instrument.

3.2 Market data: 81.0% becomes 54.0%

split scheme	balanced_acc	macro_F1	folds
most-frequent baseline	50.0%	--	--
stratified baseline	50.7%	--	--
random (optimistic)	81.0% (+/-0.4)	81.8%	5
unseen-producer	54.0% (+/-5.6)	50.3%	5
unseen-lab	68.3% (+/-18.6)	67.6%	8
dual-class + unseen	54.6% (+/-2.6)	50.7%	5

Under the protocol used in the published literature, the chemistry appears to separate solventless from hydrocarbon extracts at 81.0%. Holding out the producer leaves 54.0%, four points above coin-flipping.

Two qualifications are needed before that 27-point gap can be read as a pure protocol effect, and both are measurable.

First, the two schemes do not run on the same rows: `random` uses all 37,344 usable rows while `unseen-producer` requires a producer and so is confined to the 33,229 rows that have one, 89.0%. Comparing like with like, a random split *restricted to those same rows* scores 78.5%, so the protocol-only gap is **24.4 points** and the remaining 2.5 come from the easier sample. The paper's headline figure of 27 is the comparison a reader of the published literature would make, since those studies report the random-split number on all their data; 24.4 is the stricter quantity and we report both.

Second, the grouped result is not distinguishable from chance by a conventional test. Across the five producer-held-out folds the scores are 48.6, 57.6, 46.7, 55.7 and 61.5, giving a 95% confidence interval of [46.3, 61.8] that contains 50.0% (one-sample t against chance, $p = 0.22$). The honest reading is therefore not "54.0% is a small real effect" but "the producer-held-out performance is statistically indistinguishable from guessing". That strengthens rather than weakens the conclusion: the collapse goes all the way to chance.

The gap is stable under reseeding. Three further seeds give protocol-only gaps of 24.6, 24.5 and 24.6 points, with the grouped score at 54.0% in every case.

The strictest test agrees: 54.6% on the dual-class subset, with a tight fold spread of 2.6 points. Macro F1 is 50.3% and 50.7% on the two group-held-out schemes, against a 50.0% chance level, with fold spreads of 5.6 and 2.6. No gate that requires beating chance by more than the between-fold spread is passed.

The confusion matrices show where the apparent skill was concentrated:

unseen-producer	true hydrocarbon: 16,149 correct / 5,818 wrong
	true solventless: 3,332 correct / 7,930 wrong

Solventless recall is **32.3%**. The model mostly answers with the majority class and its residual skill lives on the easy side. A single balanced-accuracy number would have concealed this; per-class recall does not.

The `unseen-lab` scheme deserves separate comment. Its mean, 68.3%, is the highest of the three honest schemes, but its fold-to-fold standard deviation is **18.6 points**. Performance swings wildly with which laboratory is held out. That is the signature of a model keyed to laboratory convention rather than to a stable chemical difference, and it is consistent with the alarm in 3.1. A study reporting only the mean of this scheme would claim 68.3% for something that is not reproducible across laboratories.

The per-laboratory class balance shows why the spread is so large. Across the eight held-out laboratories the hydrocarbon share ranges from 13.5% (Cannalytics RI, 758 rows) to 78.4% (PureVita, 3,137 rows), against an overall 65.8%. Holding out a laboratory therefore shifts the test-set prior as well as the instrument, and the two effects cannot be separated with this data. We report the scheme for completeness and do not rest any conclusion on it.

3.3 Controlled experiment: the same collapse

split scheme	balanced_acc	macro_F1	folds
chance level (3 classes)	33.3%	--	--
random 5-fold	56.7% (+/-9.2)	55.7%	5
leave-one-variety-out	38.9% (+/-12.6)	35.9%	6

A random split suggests method is somewhat separable at 56.7%. Holding out an entire variety gives **38.9%**, 5.6 points above chance with a fold spread of 12.6 that straddles it. The same collapse as in the market data, with the confounder changed from producer to variety.

And the same inversion:

```
chemistry -> variety: balanced_acc 60.7% (+/-8.3) chance 16.7%
```

The four cannabinoids predict the **variety** at nearly four times chance, far better than they predict the extraction method they were measured to compare. Even in an experiment built specifically to isolate method, with fixed genetics per group and clean HPLC numbers, genetics dominates the cannabinoid signal.

This matters for interpretation. The Cannlytics result alone could be dismissed as a consequence of messy market data. The controlled experiment removes that explanation.

4. Discussion

4.1 What this establishes

Two claims, of different kinds.

The **chemical claim** is narrow: the analyte panels present in public cannabis COAs, 19 analytes and heavily right-censored, do not carry enough information to identify extraction method for a producer the model has never seen. Strain and producer variation dominate whatever method signal exists.

The **methodological claim** is general and, we think, the more useful one. Cannabis chemical data has group structure, and the field's standard validation protocol ignores it. Two independent datasets, one market and one designed, both show double-digit collapses when the group is held out. In both, the chemistry predicts the confounder better than the target.

This gives a concrete magnitude to the concern raised abstractly in [8]: in this domain the inflation is 27 points on market data, 24.4 of which survive a same-sample comparison, and 18 on a controlled experiment, not a rounding error.

4.2 What this does not establish

Not that extraction method leaves no chemical trace. Controlled studies with fixed genetics and fixed instrumentation measure method effects directly; this work says nothing about those.

Not that a better model would fail. Only that these features do not generalise across producers. A richer assay, a full terpene panel or an LC-MS fingerprint, is a different experiment.

Not that the labels are wrong. They are source-declared and agree with the name-derived path on 99.3% of rows where both exist.

Not sensitive to the 64 pre-roll rows. Removing them moves random-split balanced accuracy from 81.0% to 80.6% and leaves unseen-producer unchanged at 54.0%, so the 27-point gap becomes 26.6. The conclusion does not depend on their inclusion.

Not a claim about the correctness of the conclusions in [1], [2] or [3]. Their chemical findings may well hold. What we show is that the reported performance figures are obtained in a regime where we measure substantial inflation, and so cannot be read as estimates of performance on a new producer, genotype or laboratory.

4.3 The confounder we could not test

Strain is the obvious remaining candidate. Solventless production skews toward cultivars that wash well, so the class label may partly encode which cultivar was used, and cultivar drives terpene profile directly.

We could not test it. The `strain_name` and `strain_type` columns exist in the source file but are empty for **0 of 37,374** labelled rows. The only remaining cultivar signal is inside `product_name`, and extracting a strain key from it yields 37.1% obvious non-cultivars: `oil`, `thc`, `lab sample`, `ethanol`, `acres`. A probe built on a key that is one-third noise cannot distinguish "cultivar leaks" from "my regex leaks".

We report this as a data-availability limit, not a finding. For completeness, the run on the noisy key gave a *negative* lift for predicting the key from chemistry, minus 3.0 points, which if anything hints that strain is a weaker fingerprint here than laboratory was. We do not rely on it.

4.4 Recommendations

For anyone fitting a classifier to cannabis chemical data:

1. **Group by the entity production will hand you fresh.** Producer, genotype, laboratory. `GroupKFold` and `LeaveOneGroupOut` cost nothing to run.
2. **Report the random-split score alongside the grouped one.** The gap is the informative quantity and is cheap to obtain.
3. **Never report plain accuracy on an imbalanced class.** Reference [1]'s 95.2% sits on a 78.8% majority baseline, so the lift is 16.4 points.

4. **Report per-class recall and the between-fold spread.** Our `unseen-lab` mean of 68.3% carries a standard deviation of 18.6; the mean alone is misleading.
5. **Run an alarm model before the real task.** If the chemistry predicts the laboratory, any laboratory-sharing score is partly instrumental.
6. **Keep a tested-versus-detected distinction in censored panels.** Merging a measured zero with a missing cell changes the coverage gap in our data from roughly 5 points to roughly 22.

4.5 Where a decisive answer would come from

The remaining untested avenue is a method-comparison dataset pairing extraction method with a full metabolomic fingerprint including terpenes, such as the MassIVE LC-MS/GC-MS collections. That alone could distinguish "method leaves no usable trace" from "method leaves a trace, but not in the 4-cannabinoid or 19-analyte projections tested here". It is a mass-spectrometry corpus of several gigabytes and a separate build, not an extension of this work.

5. Conclusion

Extraction method is not identifiable from the cannabinoid and terpene panels in public cannabis certificates of analysis for an unseen producer: 54.0% balanced accuracy whose 95% confidence interval contains the 50.0% chance level, and 54.6% on the strictest subset, where a random split reports 81.0%. The same collapse appears in a designed HPLC experiment when variety is held out, 56.7% to 38.9% against 33.3% chance. In both datasets the chemistry reads the confounder better than the target, identifying the laboratory 15.3 points above baseline and the variety at nearly four times chance.

The field's published accuracies, 95% and 100%, are obtained under validation that leaves the group structure intact, the family of protocols for which we measure 27 points of inflation on market data, 24.4 of them on a same-sample comparison, and 18 on a designed experiment. We ran random cross-validation, not Kennard-Stone selection, so the magnitude we report is specific to the former; what the two share is that neither holds out the producer, the genotype or the laboratory. We do not claim the published chemical conclusions are wrong, and we did not rerun those studies. We claim their numbers do not answer the question a reader will assume they answer. Group-aware validation is not a refinement here. It changes the conclusion.

Data and code availability

Source data are public and are not redistributed here. The Cannlytics `cannabis_results` corpus [9] and Zenodo record 13823859 [10] are both CC-BY-4.0 and are retrieved by the ingestion scripts described below, which verify the downloaded size against the figure declared by the host API.

All analysis code and the raw evidence log for every run described in this manuscript are public at <https://github.com/ArtBreguez/cannabis-extraction-ds> (code MIT, manuscript CC-BY-4.0). The structure is:

<code>src/ingestion/</code>	retrieval with download-integrity verification
<code>src/preprocessing/</code>	labelling, unit checks, non-detect encoding
<code>src/models/</code>	the four split schemes and the alarm probes
<code>src/evaluation/</code>	recomputation and audit of every figure reported here
<code>docs/evidence/</code>	raw stdout of each run, committed unedited

A single script, `src/evaluation/rederive_paper_numbers.py`, recomputes every dataset figure in this manuscript directly from the labelled dataset, and `src/evaluation/audit_preprint.py` checks each claimed number in this text against either that recomputation or the committed log, failing on any disagreement.

Competing interests

The author produces cannabis-related content on social media under the handle @HiddenTerps and has a commercial interest in the solventless extract category. This work reports a negative result about distinguishing that category from hydrocarbon extracts, which runs against that interest rather than supporting it. No funding was received.

A note on provenance

An early version of this dataset audit concluded the research was not viable, reporting 6,336 rows and 24 versus 11 labellable samples. That measurement was made on a download truncated to 7% of its size, 177.9 MB of 2,534.8 MB, cut mid-record. `curl` reported success, the parser opened the file and discarded a single malformed line, and every downstream number was plausible. The truncation was found by comparing the local file size against the figure declared by the host API. Every number in this manuscript comes from the verified intact file, and download integrity is checked against the source in `src/ingestion/`.

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